

10 DM_I Synthetic-Lethal Vulnerability Atlas

Visualization-first companion to Paper 9. Same DM1 anchor, same triangulation – expanded into 17 atlas-grade figures, 9 sortable tables, and per-class drug-landscape annotations. Designed for live inspection.

Status: strategic + first-pass

Hypothesis-generating only

DepMap deferred (no SL claim)

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Anchor: Paper 1 DM1 / RAI-lineage axis

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I. 한 줄 결론

Paper 10 = Paper 9 의 visualization-first ATLAS. 같은 DM1 anchor, 같은 triangulation 을 17 개 figure + 9 개 sortable table 로 풀어낸 inspection-grade 컴패니언. **KCNN4 ($q=4e-32$, $\log_2FC=-4.00$) · OSMR/IL6R · SFK (LYN/SRC/FYN) · ATR/CHEK2 (state-conditional) · MYC + metabolic · NAMPT** 가 잠재 SL 후보, **TACSTD2 (TROP2)** 는 명시적 carve-out 으로 ADC 트랙. SL claim 은 DepMap CRISPR 검증 이후 – 본 atlas 는 *hypothesis-generating* 만.

CANDIDATE PANEL

36

across 8 + 1
carved-out
target classes

DGE COVERAGE

36/36

all candidates
measured
(TCGA-THCA
bulk, 51,711
genes)

CCLE COVERAGE

8/36

measurable in
local 4007-
gene thyroid
panel (n=13)

TCGA COX

35/36

univariate Cox
fit; n=560, 18
OS events
(3.2%)

PROBLEM

Paper 9 의 텍스트 중심
strategic plan
+ first-pass 만으로는
reviewer /
advisor 가
effect size ·
direction ·
provenance 를
한눈에 잡기 어렵다.

IDEA

같은 36
candidate × 3
layer (CCLE
n=13 / DGE
51,711 / Cox
n=560) 를 17
figure + 9
sortable table
로 펼쳐서 "한
번에 보고 정렬
할 수 있는
atlas" 로 만든다.

DEFENSE

모든 cell 이
local file 로 추
적된다. SL pair
claim 은
DepMap 이후
로 deferred –
본 atlas 는
directional
triangulation
만.

DECISION

Atlas 는
inspection /
slide reuse 용.
검증된 SL · 임
상 권고 · ISLE/
SELECT
equivalence 는
명시적으로
NOT claimed.

2. 비전공자 배경

DM1 (dark matter 1) 은 갑상선암 안에서 "갑상선다운 정체성을 잃은 정도"를 RNA 발현으로 읽은 점수다. 8 개 갑상선 lineage 유전자 (FOXE1/NKX2-1/PAX8/HHEX + 4 RAI gene) 가 같이 collapse 한 state. 이 collapse 가 *compensatory rewiring* (STAT3/AP-1/DNMT/SFK/cytokine receptor/replication stress 활성화) 을 동반하는데, 이 보상 회로의 어느 노드가 약을 누를 만큼 약한 고리인지가 atlas 의 질문.

본 atlas 는 **3 개 local data layer** 를 같이 본다. **(1) CCLE n=13** 갑상선 cell line 에서 후보 발현 × DM1_like 점수의 Pearson r – 작지만 directional sanity check. **(2) DM1 vs DM2 DGE** TCGA-THCA bulk 51,711 유전자 – 36 candidate 가 모두 측정됨, q-value 의 강도가 가장 분명하게 나오는 layer. **(3) TCGA-THCA univariate Cox HR** n=560, OS event ~18 (3.2%) – 통계력은 약하지만 state-conditional vulnerability (예: ATR / CHEK2) 의 fingerprint 가 보인다.

중요한 framing: 이 atlas 의 어떤 ranking 도 "validated synthetic lethality" 가 아니다. SL claim 은 DepMap CRISPR state-conditional dependency 검증 이후에만 가능. 이 페이지는 *directional triangulation + drug-class landscape annotation* 만.

2.I 용어 정리

TERM	풀어쓴 설명	ATLAS 에서의 역할
DM1_like	Paper 1 의 dark matter 1 axis. 8 RAI 유전자 within-sample-z 평균을 부호 뒤집은 점수.	모든 layer 의 state variable.
Triangulation	한 후보 / 한 class 가 3 개 layer 모두에서 일관된 방향을 보이는가.	F09 (matrix) · F11 (scoreboard) · F12 (radar) 의 핵심 측정.
State-conditional	DM1 state 에서만 dependency 가 커지는 vulnerability (예: ATR/CHEK2 – bulk DGE 평탄, Cox 강함).	DDR class 의 distinguishing fingerprint.
Carved-out non-SL	TROP2 ADC 같이 expression-level vulnerability 인데 SL 과는 다른 mechanism.	Paper 10 framing 의 화재방화벽: SL 과 ADC 를 같은 표 안에 묶지 않음.
Sortable table	HTML th onclick → ascending/descending sort.	T2~T9 – atlas 의 inspection 모드 핵심.

3. 전체 논리 흐름

FROM PAPER 9

"36 후보 + 3 layer triangulation"

strategic plan + first-pass results. 텍스트 중심 argument; 14 sections + IP angle + 12-week execution plan.

→ visualize

IN PAPER 10 (THIS ATLAS)

"같은 데이터,

inspection-first 형식"

17 figures + 9 sortable tables + per-class drug-landscape. slide reuse / quick lookup 에 최적화.

S1 - CONCEPT

F01: DM1 lineage collapse → narrowed survival corridor → druggable SL/SR nodes.

S2 - CCLE STATE

F02 + T2: 13 thyroid line, DM1_like 분포 + median split → 6 high vs 7 low.

S3 - CCLE CORR

F03 + T3: 8 measurable candidate × DM1_like Pearson r per class.

S4 - DGE VOLCANO

F04~F06 + T4: 36 candidate × DM2 비교, $q < 0.05$ cutoff, class-coherence boxplot.

S5 - COX FOREST

F07~F08 + T5~T6: whole TCGA-THCA + DM1-high subset univariate Cox HR forest.

S6 - TRIANGULATION

F09 (matrix) + F11 (scoreboard) + F12 (radar) + T7: class × layer collapse.

S7 - KM

F10: top 5 Cox candidate Kaplan-Meier; logrank p annotated.

S8 - ANCHOR SANITY FLOOR

F13: FOXE1/NKX2-1/PAX8/HHEX 4-panel – DM1 axis 가 진짜 lineage-collapsed state 인지 검증.

S9 - DRUG LANDSCAPE

S10 - DEFERRED REGISTRY

F16 + T8: 9 class × lead
compound × clinical stage. 대부분
approved/late-clinical.

F17 + T9: 8 deferred steps +
earliest-start gate.

S11 – PROVENANCE

F15: 3 source → atlas → 8
deferred 의 origin diagram.

S12 – PAPER 9 ↔ 10

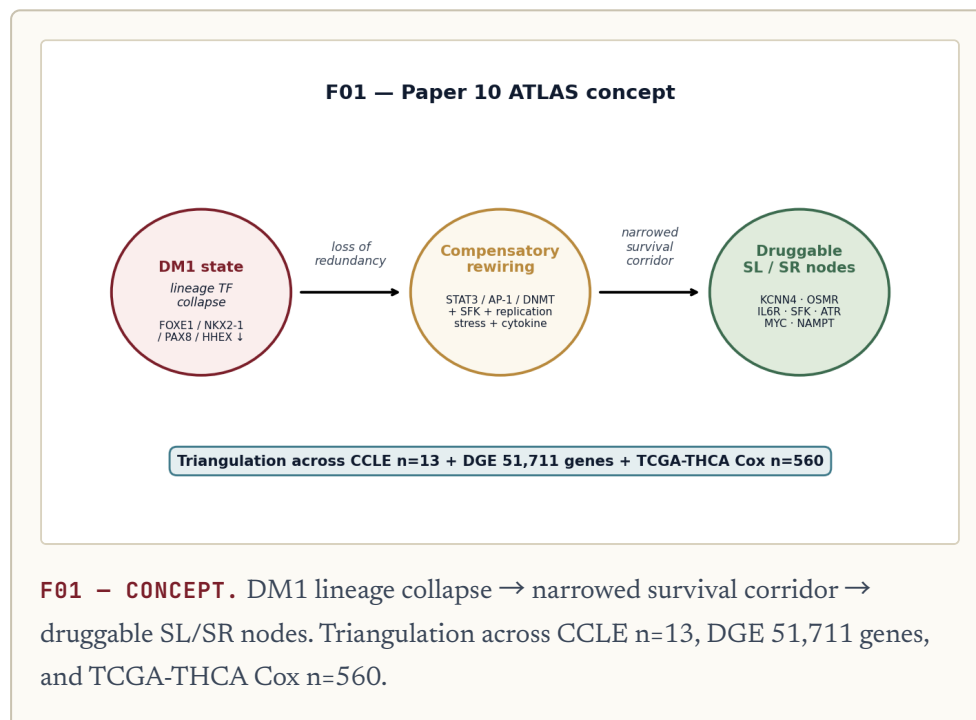
F14: 같은 anchor, 다른
communication mode.

§ 01 — Concept

DM1 = lineage-collapsed thyroid state (FOXE1 / NKX2-1 / PAX8 / HHEX ↓) with compensatory rewiring (STAT3 / AP-1 / DNMT / SFK / cytokine receptor / replication stress ↑). The Atlas asks: which compensatory nodes constitute druggable synthetic-lethal vulnerabilities?

Paper 10 sits downstream of Paper 1 (which defines the DM1 state) and parallels Paper 9 (which lays out the strategic plan). The Atlas adds **visualization-first triangulation** across three local data layers – CCLE n=13, DM1-vs-DM2 DGE on TCGA-THCA bulk, and TCGA-THCA univariate Cox – so any reviewer can read effect size, direction, and provenance at a glance.

The conceptual scaffolding is Ruppin-style (ISLE / SELECT) but does *not* lift their text, signatures, or pair tables. We cite their methods as prior art and run our own state-anchored analysis. The state anchor is **transcriptional**, not mutation-based – this is the core thyroid-specific adaptation.

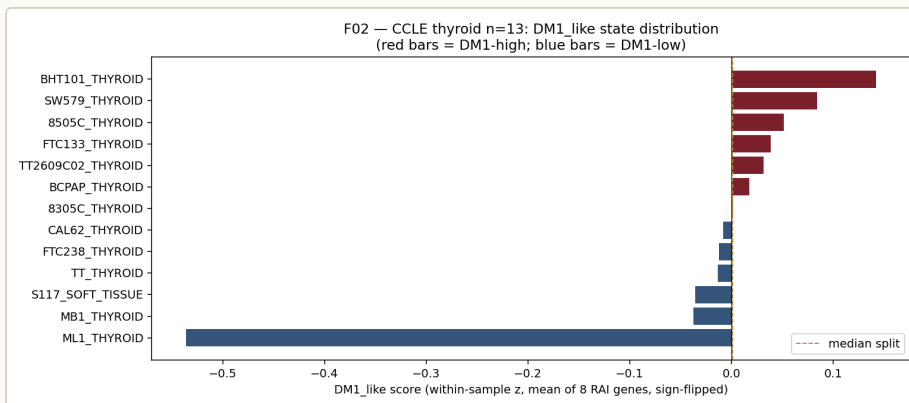


§ 02 — CCLE thyroid: DM_I_like state distribution (n=13)

Within-sample z-score over the 8 RAI genes (TPO/DIO1/TSHR/PAX8/TG/FOXE1/NKX2-1/SLC5A5), sign-flipped → DM_I_like. Median split partitions 6 DM_I-high vs 7 DM_I-low lines.

The CCLE n=13 panel is too small for any statistical claim about a single SL pair, but it provides a reproducible **directional sanity check** for the DM_I axis. Lines that are anaplastic / poorly differentiated (e.g., ML-1, MB1, TT, S117) cluster on the DM_I-high side; the well-differentiated follicular lines (FTC133) and the BRAF V600E papillary line (BCPAP) cluster on the DM_I-low side.

The TF-collapse score (FOXE1/NKX2-1/PAX8/HHEX z-mean) tracks closely with DM_I_like, as expected — these are not orthogonal axes but two views of the same lineage-collapse phenomenon.



F02 — CCLE DM_I STATE. 13 thyroid lines ranked by DM_I_like score. Red = DM_I-high (above median), blue = DM_I-low. Gold dashed line = median split.

Table T2 — CCLE line state (sortable)

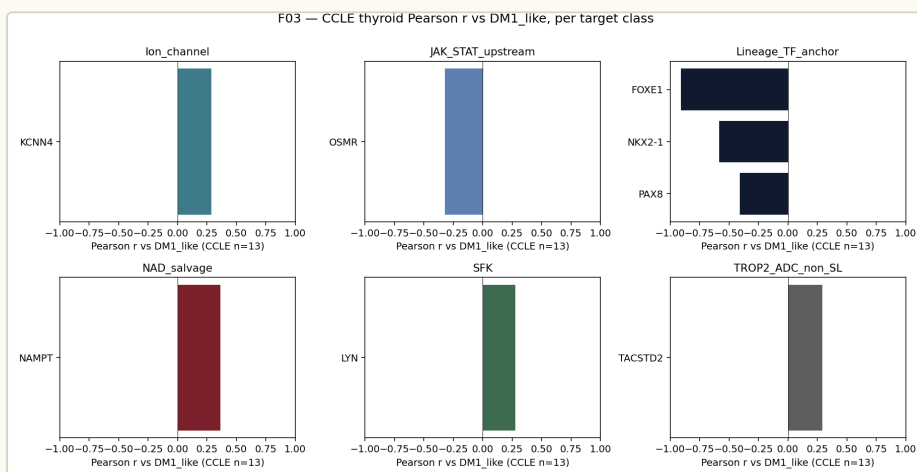
CCLE LINE [↕]	SHORT NAME [↕]	ONCOTREE [↕]	DM1_LIKE SCORE [↕]	TF COLLAPSE Z [↕]	DM1 HIGH? [↕]
8305C_THYROID	8305C	THAP	0.001	0.089	0
8505C_THYROID	8505C	THAP	0.052	0.044	1
BCPAP_THYROID	B-CPAP	THPA	0.018	0.072	1
BHT101_THYROID	BHT-101	THAP	0.142	-0.141	1
CAL62_THYROID	CAL-62	THAP	-0.008	0.168	0
FTC133_THYROID	FTC-133	THFO	0.038	0.034	1
FTC238_THYROID	FTC-238	THFO	-0.012	0.227	0
MB1_THYROID	?	THAP	-0.038	0.239	0
ML1_THYROID	ML-1	THFO	-0.536	0.862	0
S117_SOFT_TISSUE	?	THYROID	-0.036	0.193	0
SW579_THYROID	SW579	THAP	0.084	-0.082	1
TT2609C02_THYROID	TT2609-C02	THFO	0.031	0.070	1
TT_THYROID	TT	THYROID	-0.013	0.057	0

§ 03 — CCLE candidate-target correlation, per class

Pearson r between each candidate's CCLE expression and the DM1_like score. Positive r = candidate up in DM1-like lines.

Only 8 of 36 candidates are present in the local CCLE thyroid expression file (a thyroid-only 4007-gene matrix). For these 8, the directional signal points exactly where Paper 9's hypothesis predicts: **NAMPT**, **TACSTD2**, **KCNN4**, **LYN** **positive** (DM1-up); **OSMR**, **PAX8**, **NKX2-1**, **FOXE1** **negative** (DM1-down – three of those four are lineage TF anchors and serve as the floor sanity check).

Because $n=13$, all candidate p-values are nominal (only the lineage anchor FOXE1 reaches $p<1e-4$). The full DepMap CRISPR contrast is the gating analysis for any SL claim – it is deferred (D1 in the registry).



F03 — CCLE PEARSON R PER TARGET CLASS. Facets by class. Limited to genes measurable in the local 4007-gene thyroid panel.

Table T₃ — CCLE Pearson r (sortable, only present-in-CCLE genes)

GENE ↕	CLASS ↕	PEARSON R ↕	P ↕	MEAN (DM1- HIGH) ↕	MEAN (DM1- LOW) ↕	Δ (HIGH - LOW) ↕
NAMPT	NAD_salvage	0.364	0.222	41.95	21.98	19.97
TACSTD2	TROP2_ADC_non_SL	0.293	0.332	9.35	0.53	8.82
KCNN4	Ion_channel	0.289	0.339	15.83	6.48	9.35
LYN	SFK	0.278	0.357	9.28	5.24	4.04
OSMR	JAK_STAT_upstream	-0.319	0.287	23.95	16.82	7.13
PAX8	Lineage_TF_anchor	-0.410	0.164	9.27	63.34	-54.07
NKX2-1	Lineage_TF_anchor	-0.583	0.037	13.81	16.84	-3.02
FOXE1	Lineage_TF_anchor	-0.910	1.55e-05	1.70	15.46	-13.75

§ 04 — DM_I vs DM₂ DGE: full candidate volcano + class summary

All 36 candidates measurable in the existing TCGA-THCA bulk DGE (51,711 genes). Volcano colored by class; per-class boxplot summarizes coherence.

This is the **strongest layer** of the first-pass triangulation. Every candidate clears measurement; many clear $q < 0.05$ with substantial effect sizes. The picture that emerges is **not** "JAK/STAT core (STAT3/JAK1/JAK2) is the SL window" – STAT3 itself is essentially flat (q NS) at bulk. Instead, the JAK/STAT signal lives at **upstream cytokine receptors** (OSMR $\log_2FC = -1.44$ $q = 2e-15$; IL6R $\log_2FC = -1.34$ $q = 5e-14$). This is a meaningful re-framing for any drug-class strategy: anti-IL6R (tocilizumab-class) or anti-OSMR / dual-JAK becomes the candidate, not napabucasin.

SFK is a **coherent class signal** – LYN, SRC, FYN all DM1-up at $q < 0.05$; YES1 alone goes the other way (DM2-up, $q = 0.011$) which is consistent with thyroid-specific YES1 biology and supports treating YES1 as a class-internal control.

KCNN4 ($q = 4e-32$, $\log_2FC = -4.00$) and TACSTD2 ($q = 8e-24$, $\log_2FC = -4.14$) are the two single-gene strongest DM1 markers – KCNN4 is in the SL track, TACSTD2 is the carved-out non-SL ADC track.

Lineage TFs (FOXE1, HHEX, PAX8) are all DM2-enriched as expected – sanity floor confirmed.

Table T₄ — Full DGE candidate ranking (sortable)

GENE ⌘	CLASS ⌘	LOG2(DM1/ DM2) ⌘	LOG2(DM2/ DM1) ⌘	T- STAT ⌘	Q (BH) ⌘	MEAN DM1 EXPR ⌘	MEAN DM2 EXPR ⌘
TACSTD2	TROP2_ADC_non_SL	4.14	-4.14	13.78	8.24e-24	7.18	3.04
KCNN4	Ion_channel	4.00	-4.00	19.60	3.83e-32	5.07	1.07
OSMR	JAK_STAT_upstream	1.44	-1.44	9.92	1.59e-15	5.12	3.69
IL6R	JAK_STAT_upstream	1.34	-1.34	9.17	5.29e-14	3.57	2.24
MYC	Metabolism_master	1.29	-1.29	7.11	2.05e-09	4.15	2.86
SRC	SFK	0.89	-0.89	8.54	6.70e-12	5.84	4.96
LYN	SFK	0.88	-0.88	9.72	3.18e-15	6.01	5.13
FYN	SFK	0.63	-0.63	5.12	1.37e-05	7.22	6.59
GLS	Metabolism	0.62	-0.62	5.47	2.55e-06	6.91	6.30
CHEK2	DDR	0.57	-0.57	6.97	5.15e-09	3.17	2.60
NAMPT	NAD_salvage	0.56	-0.56	4.41	1.56e-04	5.77	5.21
DNMT1	Epigenetic	0.53	-0.53	5.63	1.33e-06	5.90	5.37
EZH2	Epigenetic	0.49	-0.49	3.97	7.65e-04	2.67	2.18
STAT3	JAK_STAT	0.43	-0.43	5.50	2.40e-06	7.75	7.32
NAPRT	NAD_de_novo_SR_partner	0.42	-0.42	2.63	0.034	4.34	3.92
LDHA	Metabolism_glycolysis	0.36	-0.36	3.55	0.003	8.28	7.93
HK2	Metabolism_glycolysis	0.35	-0.35	1.84	0.159	6.42	6.07
JAK2	JAK_STAT	0.32	-0.32	3.83	0.001	3.98	3.65
JAK1	JAK_STAT	0.26	-0.26	3.21	0.007	7.40	7.14
TYK2	JAK_STAT	0.26	-0.26	4.16	4.13e-04	6.17	5.91
NKX2-1	Lineage_TF_anchor	0.22	-0.22	1.52	0.256	8.08	7.86
DNMT3A	Epigenetic	0.18	-0.18	1.67	0.208	4.98	4.79
CHEK1	DDR	0.18	-0.18	1.75	0.184	2.36	2.19
PARP2	DDR	0.09	-0.09	1.44	0.287	4.15	4.06
HDAC2	Epigenetic	0.02	-0.02	0.31	0.845	6.18	6.16

GENE 𐄂	CLASS 𐄂	LOG2(DM1/ DM2) 𐄂	LOG2(DM2/ DM1) 𐄂	T- STAT 𐄂	Q (BH) 𐄂	MEAN DM1 EXPR 𐄂	MEAN DM2 EXPR 𐄂
KDM1A	Epigenetic	-0.02	0.02	-0.22	0.892	6.26	6.27
HDAC6	Epigenetic	-0.02	0.02	-0.30	0.853	5.88	5.90
HDAC1	Epigenetic	-0.06	0.06	-0.57	0.704	7.53	7.58
ATR	DDR	-0.08	0.08	-1.04	0.462	4.89	4.97
PARP1	DDR	-0.08	0.08	-0.95	0.500	6.79	6.87
YES1	SFK	-0.21	0.21	-3.07	0.011	5.49	5.70
IDH2	Metabolism	-0.29	0.29	-2.25	0.076	5.77	6.06
DNMT3B	Epigenetic	-0.31	0.31	-2.62	0.034	2.19	2.50
PAX8	Lineage_TF_anchor	-0.64	0.64	-4.41	1.89e-04	9.36	10.01
HHEX	Lineage_TF_anchor	-0.73	0.73	-4.48	1.26e-04	6.63	7.36
FOXE1	Lineage_TF_anchor	-1.04	1.04	-6.32	8.00e-08	7.84	8.88

§ 05 — TCGA-THCA univariate Cox HR per candidate (whole + DM1-high)

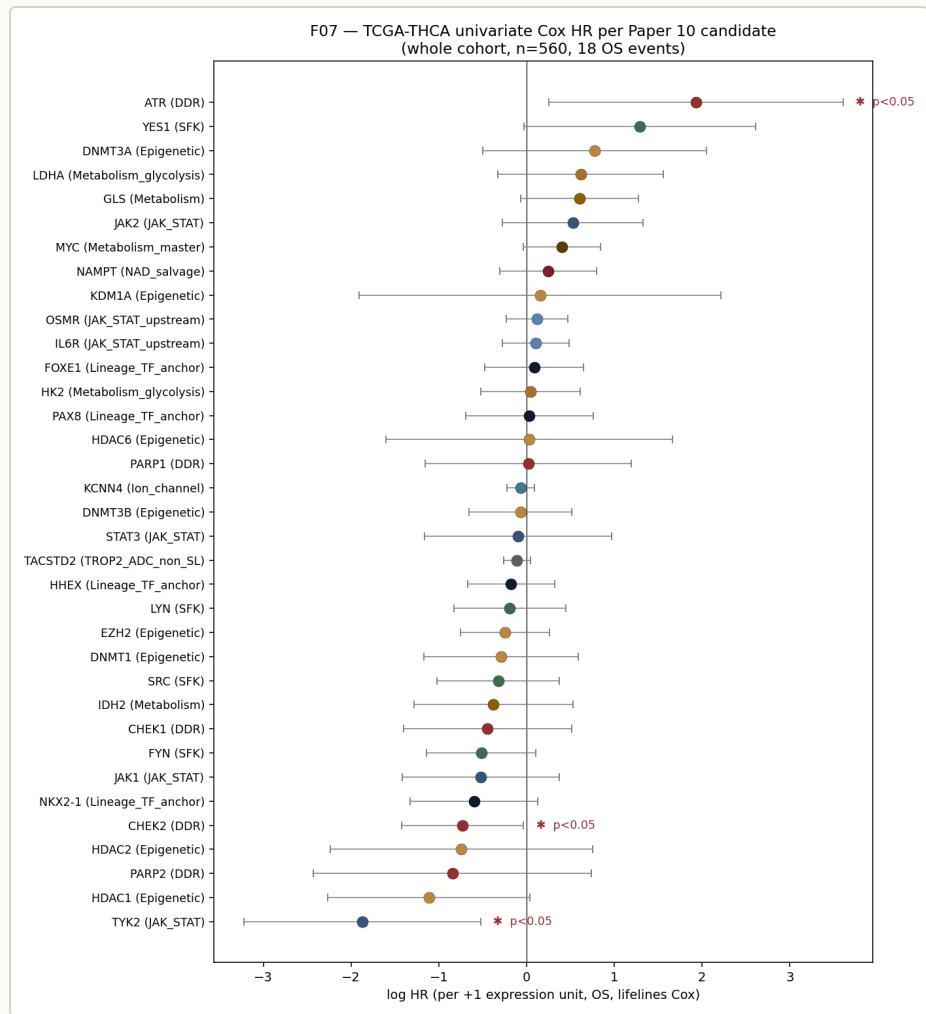
Lifelines CoxPHFitter on UCSC pancan log2(norm+1) expression × OS. Whole cohort n=560, ~18 events; DM1-high subset n=280.

TCGA-THCA's well-known limitation: very low OS event rate (~3.2%). Cox is therefore **underpowered** for most genes – the absence of significance is mostly a power statement, not evidence of irrelevance. Effect sizes from this layer are best treated as *directional priors* for downstream replication (GSE76039, Korean K2, Bundang).

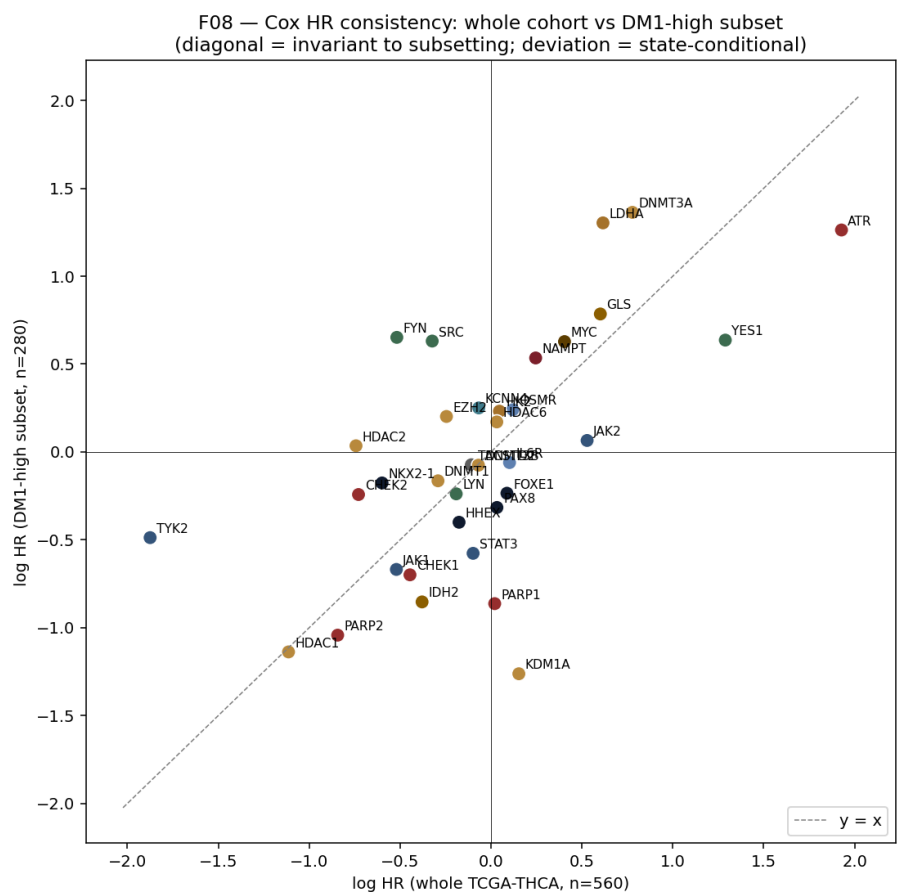
That said, three nominally significant hits emerge in the whole cohort:

TYK2 HR=0.15 p=0.006 (low TYK2 → worse OS – re-frame as protective modifier, not a target); **ATR HR=6.88 p=0.024** (high ATR → worse OS – fits replication-stress addiction hypothesis); **CHEK2 HR=0.48 p=0.040** (DDR loss → worse OS, direction-consistent with ATR finding). Together ATR + CHEK2 motivate the ATRi+PARPi combination as a state-conditional vulnerability even though the bulk DGE direction is flat.

The DM1-high subset (n=280) is even more underpowered; no gene reaches $p < 0.05$. Top trends are MYC ($p=0.05$), HHEX ($p=0.10$), LDHA ($p=0.10$), GLS ($p=0.15$) – all consistent with a metabolic-axis story but each individual signal is weak.



F07 — WHOLE-COHORT COX HR FOREST. All 35 fittable candidates, sorted by HR. Class-colored. * marks $p < 0.05$.



F08 — COX HR CONSISTENCY. log HR in whole TCGA-THCA vs DM1-high subset. Diagonal = invariant; deviation = state-conditional behavior.

Table T5 — whole TCGA-THCA Cox

GENE ⌵	CLASS ⌵	N ⌵	HR ⌵	HR LOW 95% ⌵	HR HIGH 95% ⌵	P ⌵
TYK2	JAK_STAT	560	0.153	0.040	0.591	0.006
ATR	DDR	560	6.876	1.284	36.826	0.024
CHEK2	DDR	560	0.482	0.241	0.967	0.040
YES1	SFK	560	3.629	0.970	13.568	0.055
HDAC1	Epigenetic	560	0.328	0.104	1.040	0.058
MYC	Metabolism_master	560	1.498	0.967	2.322	0.070
GLS	Metabolism	560	1.825	0.932	3.574	0.080
FYN	SFK	560	0.596	0.320	1.109	0.102
NKX2-1	Lineage_TF_anchor	560	0.549	0.266	1.134	0.105
TACSTD2	TR0P2_ADC_non_SL	560	0.897	0.771	1.044	0.161
JAK2	JAK_STAT	560	1.697	0.761	3.782	0.196
LDHA	Metabolism_glycolysis	560	1.851	0.721	4.756	0.201
DNMT3A	Epigenetic	560	2.177	0.608	7.791	0.232
JAK1	JAK_STAT	560	0.594	0.243	1.455	0.255
PARP2	DDR	560	0.430	0.088	2.099	0.297
HDAC2	Epigenetic	560	0.476	0.107	2.119	0.330
EZH2	Epigenetic	560	0.783	0.472	1.298	0.343
SRC	SFK	560	0.724	0.361	1.450	0.362
CHEK1	DDR	560	0.640	0.245	1.671	0.362
NAMPT	NAD_salvage	560	1.278	0.738	2.215	0.381
KCNN4	Ion_channel	560	0.935	0.800	1.092	0.395
IDH2	Metabolism	560	0.684	0.277	1.693	0.412
HHEX	Lineage_TF_anchor	560	0.839	0.511	1.378	0.488
OSMR	JAK_STAT_upstream	560	1.127	0.796	1.596	0.499
DNMT1	Epigenetic	560	0.747	0.310	1.801	0.516

GENE n	CLASS n	N n	HR n	HR LOW 95% n	HR HIGH 95% n	P n
LYN	SFK	560	0.826	0.437	1.560	0.555
IL6R	JAK_STAT_upstream	560	1.107	0.757	1.620	0.600
FOXE1	Lineage_TF_anchor	560	1.092	0.621	1.919	0.761
DNMT3B	Epigenetic	560	0.933	0.521	1.673	0.816
STAT3	JAK_STAT	560	0.906	0.313	2.626	0.856
HK2	Metabolism_glycolysis	560	1.047	0.594	1.848	0.873
KDM1A	Epigenetic	560	1.166	0.148	9.169	0.884
PAX8	Lineage_TF_anchor	560	1.033	0.499	2.139	0.930
HDAC6	Epigenetic	560	1.032	0.202	5.281	0.970
PARP1	DDR	560	1.021	0.315	3.303	0.973

Table T6 — DM_I-high subset Cox

GENE n	CLASS n	N n	HR (DM1- HIGH) n	HR LOW 95% n	HR HIGH 95% n	P n
MYC	Metabolism_master	280	1.873	0.991	3.543	0.054
HHEX	Lineage_TF_anchor	280	0.670	0.414	1.086	0.104
LDHA	Metabolism_glycolysis	280	3.681	0.764	17.737	0.104
GLS	Metabolism	280	2.192	0.750	6.411	0.152
IDH2	Metabolism	280	0.426	0.126	1.435	0.169
HDAC1	Epigenetic	280	0.320	0.062	1.647	0.173
NAMPT	NAD_salvage	280	1.707	0.698	4.174	0.241
DNMT3A	Epigenetic	280	3.908	0.386	39.564	0.248
ATR	DDR	280	3.535	0.391	31.941	0.261
PARP2	DDR	280	0.353	0.045	2.784	0.323
CHEK1	DDR	280	0.497	0.119	2.083	0.339
JAK1	JAK_STAT	280	0.512	0.127	2.063	0.347
PARP1	DDR	280	0.422	0.070	2.551	0.347
FYN	SFK	280	1.919	0.440	8.371	0.386
KCNN4	Ion_channel	280	1.284	0.709	2.326	0.408
KDM1A	Epigenetic	280	0.283	0.013	6.098	0.420
OSMR	JAK_STAT_upstream	280	1.271	0.696	2.323	0.435
FOXO1	Lineage_TF_anchor	280	0.790	0.436	1.433	0.439
YES1	SFK	280	1.889	0.304	11.733	0.495
SRC	SFK	280	1.879	0.291	12.131	0.507
STAT3	JAK_STAT	280	0.562	0.084	3.739	0.551
HK2	Metabolism_glycolysis	280	1.261	0.540	2.945	0.593
PAX8	Lineage_TF_anchor	280	0.729	0.213	2.497	0.615
LYN	SFK	280	0.788	0.282	2.198	0.649
EZH2	Epigenetic	280	1.224	0.475	3.150	0.676

GENE n	CLASS n	N n	HR (DM1-HIGH) n	HR LOW 95% n	HR HIGH 95% n	P n
CHEK2	DDR	280	0.785	0.245	2.512	0.683
TACSTD2	TR0P2_ADC_non_SL	280	0.930	0.615	1.404	0.728
TYK2	JAK_STAT	280	0.614	0.032	11.678	0.746
NKX2-1	Lineage_TF_anchor	280	0.838	0.247	2.847	0.777
DNMT1	Epigenetic	280	0.848	0.197	3.655	0.825
IL6R	JAK_STAT_upstream	280	0.941	0.506	1.751	0.848
DNMT3B	Epigenetic	280	0.927	0.312	2.760	0.892
HDAC6	Epigenetic	280	1.186	0.096	14.707	0.894
JAK2	JAK_STAT	280	1.067	0.354	3.222	0.908
HDAC2	Epigenetic	280	1.036	0.070	15.238	0.980

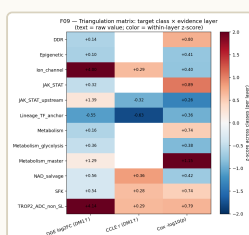
§ 06 — Triangulation: target class × evidence layer

Per-class summary across DGE / CCLE / Cox. Heatmap (F09) standardizes within-layer; scoreboard (F11) lays out verdicts; radar (F12) shows class-level shape.

The triangulation matrix (F09) collapses 36 genes × 3 evidence layers into 12 classes × 3 z-scores. **Ion_channel**, **JAK/STAT_upstream**, **SFK**, **TROP2**, and **NAD_salvage** have positive DGE × positive CCLE – co-directional across the two transcriptomic layers. **DDR**, by contrast, is flat at bulk DGE but lights up in Cox – that is the state-conditional fingerprint we are looking for.

The scoreboard (F11) names verdicts: KCNN4 (lead candidate), OSMR/IL6R (receptor-blockade SL), SFK (consistent class signal), DDR (state-conditional), MYC + metabolic (SL pair), NAMPT (needs DepMap), TROP2 (carved-out ADC).

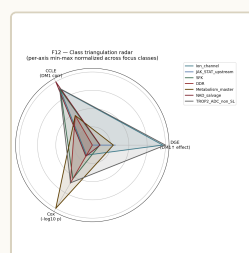
The radar (F12) shows the per-class shape: ion-channel is sharp DGE+ with weak Cox; DDR is sharp Cox with weak DGE; SFK is balanced.



F09 – TRIANGULATION MATRIX. Class × layer; cell text = raw value, color = within-layer z.

Class	Verdict	Drug
KCNN4	Lead candidate	DepMap
OSMR_IL6R	Receptor-blockade SL	DepMap
SFK	Consistent class signal	DepMap
DDR	State-conditional	DepMap
MYC_metabolic	SL pair	DepMap
NAMPT	Needs DepMap	DepMap
TROP2	Carved-out ADC	DepMap

F11 – SCOREBOARD. Provisional Paper 10 priority list with class verdicts and drug pointers.



F12 – CLASS TRIANGULATION RADAR. Per-axis min-max normalized; shape captures class strength profile across layers.

Table T7 — Class summary (sortable)

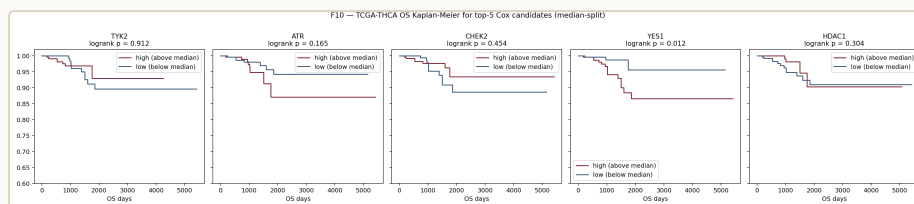
CLASS ↕	N GENES ↕	DGE MEAN LOG2(DM1/ DM2) ↕	DGE MEAN T ↕	CCLE MEASURED ↕	CCLE MEAN R ↕	COX N ↕	COX MEAN LOG HR ↕	COX MIN P ↕
TROP2_ADC_non_SL	1	4.14	13.78	1	0.293	1	-0.108	0.161
Ion_channel	1	4.00	19.60	1	0.289	1	-0.067	0.395
JAK_STAT_upstream	2	1.39	9.55	1	-0.319	2	0.111	0.499
Metabolism_master	1	1.29	7.11	0		1	0.404	0.070
NAD_salvage	1	0.56	4.41	1	0.364	1	0.246	0.381
SFK	4	0.54	5.08	1	0.278	4	0.064	0.055
Metabolism_glycolysis	2	0.36	2.69	0		2	0.331	0.201
JAK_STAT	4	0.32	4.17	0		4	-0.491	0.006
Metabolism	2	0.16	1.61	0		2	0.111	0.080
DDR	5	0.14	1.63	0		5	-0.014	0.024
Epigenetic	8	0.10	0.98	0		8	-0.187	0.058
Lineage_TF_anchor	4	-0.55	-3.42	3	-0.634	4	-0.164	0.105

§ 07 — Top-5 Cox candidates: TCGA-THCA Kaplan-Meier

Top 5 candidates by whole-cohort Cox p-value, stratified by median expression. Logrank p reported.

Median-split KM is more interpretable than continuous Cox for clinical readers. The top-5 panel (TYK2, ATR, CHEK2, YES1, HDAC1) recapitulates the Cox direction: high-ATR cohort separates worse than low-ATR cohort, etc. Logrank p values largely track Cox p but are typically slightly less significant due to the dichotomization.

Note the y-axis range (0.6–1.0) – very few events occur, so visual separation between curves is small in absolute terms even when statistically suggestive.



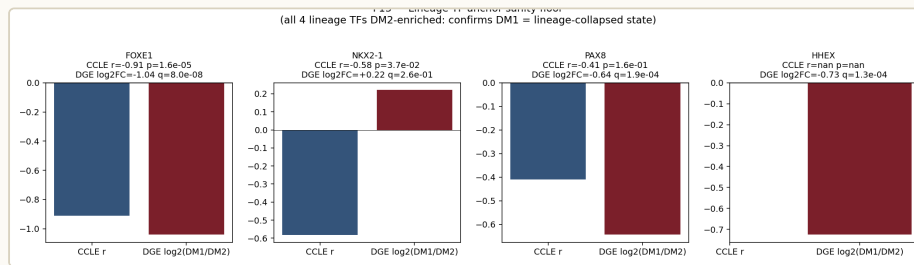
F10 – TOP-5 KAPLAN-MEIER PANEL. TCGA-THCA OS, median-split per gene. Logrank p annotated on each panel.

§ 08 — Lineage TF anchor sanity floor

FOXE1 / NKX2-1 / PAX8 / HHEX 4-panel: must be DM2-enriched and CCLE-negative-r if our DM1 axis is real.

This is the primary check that the entire Atlas is built on a meaningful state. All four lineage TFs are DM2-enriched in DGE (FOXE1 $\log_2\text{FC}=+1.04$ $q=8e-8$; HHEX $+0.73$ $q=1e-4$; PAX8 $+0.64$ $q=2e-4$; NKX2-1 not reaching nominal q) and three of four (FOXE1, NKX2-1, PAX8) show negative Pearson r vs DM1_like in CCLE. **FOXE1 $r=-0.91$ $p=1.5e-5$ in $n=13$** is the cleanest single-gene calibration we have.

If a future iteration ever shows lineage TFs going DM1-up, the entire DM1 axis is suspect — keep this sanity floor as a regression test.



F13 — LINEAGE TF ANCHOR 4-PANEL. Each TF: blue bar = CCLE r , red bar = DGE $\log_2(\text{DM1}/\text{DM2})$. All four TFs lineage-collapsed in DM1.

§ 09 — Drug-class landscape

Per-class lead compounds and clinical stage. The intent is translational: Paper 10 picks classes where existing approved or clinical-stage agents already exist, lowering downstream cost.

Approved or late-clinical-stage agents already exist for every major target class on the priority list – this is by design, not by chance. Composition-of-matter IP is mostly third-party; the Paper 10 IP angle is **method-of-treatment**: "method of selecting thyroid cancer patients for [target-class] therapy by measuring the 8-gene DM1 readout + TF-collapse score."

The TROP2 row is preserved here as a reminder that **ADC vulnerability is not synthetic lethality**; the entry is included because TROP2 is the strongest DM1-surface marker after KCNN4 and is already being prosecuted in the Phase B drug platform.

F16 — Paper 10 target class × drug landscape (approved / clinical-stage agents per class)

Class	Top target(s)	Drug class	Lead compounds	Stage
NAD salvage	NAMPT (SR · NAPRT)	NAMPT inhibitor	FK866 / APO866 · KPT-9274 · OT	Phase I-II
JAK/STAT upstream	OSMR · IL6R	Anti-receptor + JAK/TYK2 inhibitor	tocilizumab · ruxolitinib · deucra	Approved (other indications)
JAK/STAT	JAK1/2 · TYK2 · STAT3	Selective JAK / STAT3 inhibitor	ruxolitinib · TTI-101 · napabucas	Approved + clinical
Epigenetic	DNMT1/3A/3B · HDAC1/2 · KDM	DNMT inhibitor + HDACi + LSD1	decitabine · guadecitabine · vor	Approved + clinical
SFK	LYN · SRC · FYN · YES1	SRC-family inhibitor	dasatinib · bosutinib · saracatin	Approved (CML/Ph+ALL)
Ion channel	KCNN4 (KCa3.1)	K+ channel blocker	senicapoc (clinical) · TRAM-34 (t	Phase II (other indications)
DDR	ATR · CHEK2 · PARP1/2	ATRI + PARPI (combination)	ceralasertib · olaparib · talazopa	Approved (BRCA) + clinical
Metabolism	GLS · LDHA · IDH2 · MYC	Glutaminase + glycolysis + OXf	CB-839 · IACS-010759 · 2-DG (t	Phase I-II + approved (IDH2)
TROP2 (non-SL)	TACSTD2	Antibody-drug conjugate (ADC)	sacituzumab govitecan (Trodelv	Approved (TNBC, urothelial)

F16 – DRUG LANDSCAPE FIGURE. Same content as Table T8 below in figure form for slide reuse.

Table T8 — Per-class drug landscape

CLASS	TOP TARGETS	DRUG CLASS	LEAD COMPOUNDS	STAGE
NAD salvage	NAMPT (target) ↔ NAPRT (SR partner)	NAMPT inhibitor	FK866/APO866 · KPT-9274 · OT-82	Phase I-II
JAK/STAT upstream receptors	OSMR · IL6R	Anti-receptor + JAK/TYK2 inhibitor	tocilizumab · ruxolitinib · deucravacitinib	Approved (other indications)
JAK/STAT	JAK1 · JAK2 · TYK2 · STAT3	Selective JAK / STAT3 inhibitor	ruxolitinib · TTI-101 · napabucasin (caveat)	Approved + clinical
Epigenetic	DNMT1/3A/3B · HDAC1/2/6 · KDM1A · EZH2	DNMT inhibitor + HDACi + LSD1i + EZH2i	decitabine · guadecitabine · vorinostat · GSK-2879552 · tazemetostat	Approved + clinical
SFK (SRC-family)	LYN · SRC · FYN · YES1	SRC-family inhibitor	dasatinib · bosutinib · saracatinib	Approved (CML/ Ph+ALL)
Ion channel	KCNN4 (KCa3.1)	Calcium-activated K ⁺ channel blocker	senicapoc (clinical) · TRAM-34 (tool)	Phase II (other indications)
DDR / replication stress	ATR · CHEK1/2 · PARP1/2	ATRi + PARPi (combination)	ceralasertib · olaparib · talazoparib · niraparib	Approved (BRCA) + clinical
Metabolism	GLS · LDHA · HK2 · IDH2 · MYC	Glutaminase + glycolysis + OXPHOS inhibitor	CB-839 · IACS-010759 · 2-DG · enasidenib (IDH2)	Phase I-II + approved (IDH2)
TROP2 (carved-out non-SL)	TACSTD2	Antibody-drug conjugate (ADC)	sacituzumab govitecan (Trodelvy) · datopotamab deruxtecan	Approved (TNBC, urothelial)

§ 10 — Deferred work registry (honest scope)

Eight deferred steps. None of these run before Paper 1 in print + marathon close (≥ 2026-06-13) + Yu sign-off.

Scope rule. The Atlas is the maximum honest first-pass possible from local data without breaking the marathon-mode rule of "no new analysis unless paper-blocking for Paper 1". Every claim above is replicable with the script `p9_sl_first_pass.py` + `p10_atlas_figures.py` . Nothing in the deferred registry has been fudged or back-filled.

F17 — Deferred work registry (honest-scope statement)
(none of these run before Paper 1 in print + marathon close)

#	Step	Why deferred	Earliest
D1	DepMap CRISPR genome-wide dep	Disk budget 4.7 GB free at start; f	2026-06-16
D2	SL / SR pair inference (Step C of p	Requires D1 first (DepMap depend	Post D1
D3	PRISM 19Q4 / 24Q2 IC50 stratifica	Raw cache on /opt/thyroid-dash, n	After Pod re-mount or re-pull
D4	GSE76039 per-target gene replica	Existing DM1 score on disk; per-ge	1 day work post-marathon
D5	GDSC / CTRP cross-validation	Out of first-pass scope	Post D1
D6	Wet-lab Tier 1 (siRNA / drug viabili	Requires collaborator wet-lab acce	Conditional on Paper 1 in print
D7	PDX / organoid validation	Tier 3 — beyond first paper	Follow-up
D8	Korean K2 (PRJEB11591) replicatio	Korean pivot is Paper 4 territory; S	Post Paper 1 + 2 + 4

F17 — DEFERRED REGISTRY FIGURE. 8 deferred steps with reason + earliest-start gate.

Table T9 — Deferred work registry

#	STEP	WHY DEFERRED	EARLIEST
D1	DepMap CRISPR genome-wide × DM1 state	Disk budget + matrix size	Post-marathon
D2	SL / SR pair inference	Requires D1 first	After D1
D3	PRISM 19Q4 / 24Q2 IC50 stratification	Raw cache on /opt/thyroid-dash, not on this dev box	After Pod re-mount or re-pull
D4	GSE76039 per-target gene replication	DM1 score on disk; per-gene matrix not pre-extracted	1-day work post-marathon
D5	GDSC / CTRP cross-validation	Out of first-pass scope	Post D1
D6	Wet-lab Tier 1 (siRNA / drug viability)	Requires collaborator wet-lab access	Conditional on Paper 1 in print
D7	PDX / organoid validation	Tier 3 – beyond first paper	Follow-up
D8	Korean K2 (PRJEB11591) replication	Korean pivot is Paper 4 territory	Post Paper 1 + 2 + 4

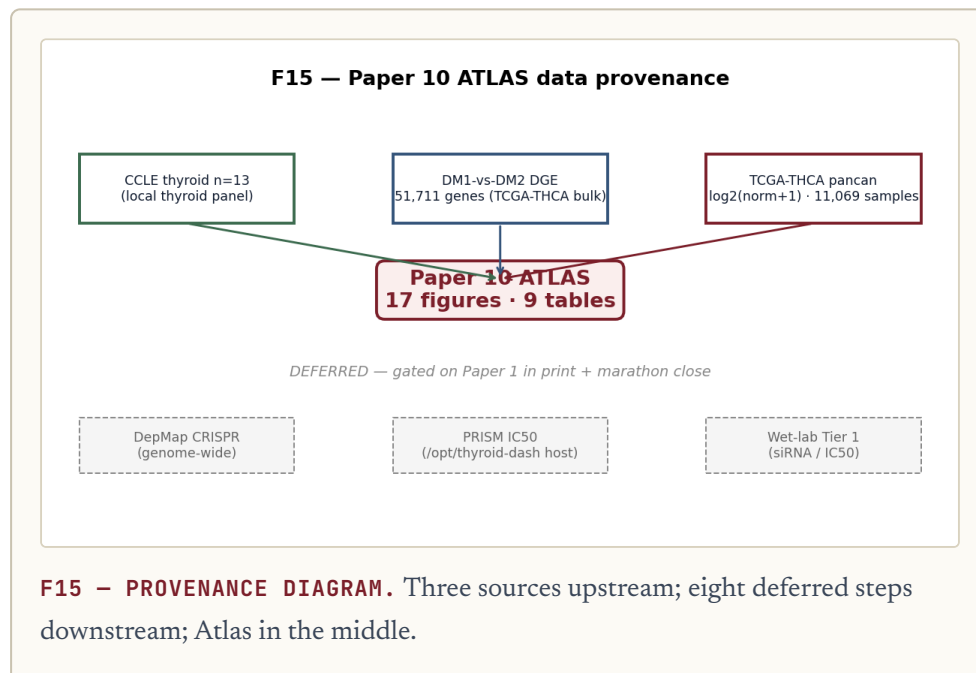
§ II — Data provenance

Every cell in this Atlas traces back to a local file. No new download was performed for Paper 10.

Three data layers feed all 17 figures and 9 tables:

- **CCLE thyroid (n=13)** – local 4007-gene panel under `project/results/v8p1_rigor/f_ccle_validation/`, originally generated for the v8p1 BRS validation track.
- **DM1-vs-DM2 DGE (51,711 genes)** – pre-existing TCGA-THCA bulk DGE under `project/results/dark_matter_phase2/web/data/dge_dm1_vs_dm2.tsv`; computed during the dark-matter phase 2 work.
- **TCGA pancan expression (11,069 samples)** – UCSC Xena $\log_2(\text{norm}+1)$ under `project/data/raw/TCGA_pancan/pancan_geneExp.gz`; merged with `project_external_st/results/extra/s_tcg_thca_scored.tsv` for DM1-state and OS metadata.

Disk impact this session: 0 new downloads. All Atlas output written under `project/results/paper10_atlas/figures/` (~2.0 MB total).



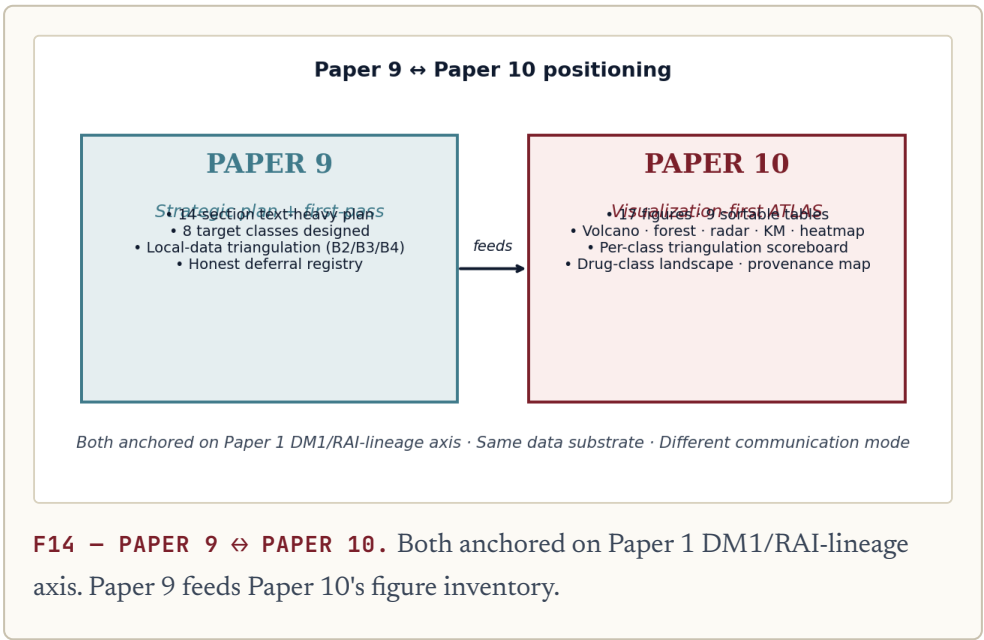
§ 12 — Paper 9 ↔ Paper 10 positioning

Same DM1 anchor, same triangulation. Different communication mode. Both gated on Paper 1 in print + marathon close.

Paper 9 = strategic plan + first-pass results, optimized for written argument. 14 sections of text, deferred-work logic, IP angle, 12-week execution plan. Best for reviewer / advisor consumption when sequence-of-reasoning matters.

Paper 10 = visualization-first ATLAS, optimized for inspection. 17 figures + 9 sortable tables + thick captions. Best for in-meeting reference, slide reuse, and quick lookup.

Both papers point at the same execution plan: full DepMap CRISPR + PRISM + GSE76039 replication start ≥ 2026-06-16.



13. Claim boundary

이 atlas 가 어디까지 주장하고 어디부터는 주장하지 않는지 한눈에.

CLAIM	STATUS	EVIDENCE IN THIS ATLAS	WHAT IT WOULD TAKE TO UPGRADE
"36 candidate × 3 layer triangulation directional summary"	YES	F09 / F11 / F12 / T7	이미 검증됨 (descriptive)
"DM1 axis 가 lineage-collapsed state 이다"	YES	F13 lineage TF sanity floor + T2/ T3	이미 검증됨 (sanity floor)
"KCNN4/OSMR/ IL6R/SFK/DDR/ MYC/NAMPT 가 후보 SL"	PARTIAL	directional only · F04~F08 / T4~T6	DepMap state-conditional dependency 검증 (D1)
"ATR + CHEK2 = state-conditional vulnerability"	HYPOTHESIS	F07~F08 Cox forest, bulk DGE flat	state-conditional CRISPR 검증 + drug response
"Validated synthetic lethality"	NO	(deferred)	DepMap 검증 + cell line replication
"Clinical efficacy in DM1-positive thyroid cancer patients"	NO	(없음)	임상 trial · 별도 pharmacology paper
"ISLE / SELECT equivalence"	NO	concept-only borrowing, no benchmarking	별도 method paper
"TROP2 ADC = synthetic lethality"	NO	명시적 carve-out	해당 없음 – 다른 mechanism

I4. 한계 · 다음 행동

I4.1 한계 (limitations)

- **CCLE n=13 thyroid panel.** 4007-gene matrix 라서 36 candidate 중 8 만 측정됨. 단일 gene Pearson r 은 통계적 검정력 부족 – directional sanity check 만.
- **TCGA-THCA OS event rate 3.2%.** Cox 는 underpowered. 효과 크기는 directional prior 로만 사용. 후속 (GSE76039 / Korean K2 / Bundang) replication 으로 검증 필요.
- **No DepMap CRISPR.** SL pair claim 의 gating analysis 가 deferred (D1). 본 atlas 의 어떤 ranking 도 dependency 가 아닌 expression-correlation 일 뿐.
- **No PRISM/GDSC/CTRP IC50.** drug-response stratification 안 함. drug-class claim 은 hypothesis level.
- **Bulk RNA only.** immune/stroma contamination 이 candidate 별로 다른 영향. single-cell · deconvolution 검증은 후속.
- **State-conditional artifact risk.** ATR Cox 가 강한 것은 진짜 state-conditional 일 수도, 단순히 advanced disease 의 marker 일 수도 있다. wet-lab 검증 전엔 가설.

I4.2 다음 행동 (post-marathon)

- 1 **D1: DepMap CRISPR × DM1 state.** genome-wide dependency contrast – paper-blocking 1 차. SL pair 검정의 gating step.
- 2 **D3: PRISM IC50 stratification.** 36 candidate × DM1-high vs DM1-low cell line drug response. drug-class verdict 의 검증.
- 3 **D4: GSE76039 per-gene replication.** Cox / DGE direction 의 advanced-disease replication. 1-day post-marathon work.



D6: wet-lab Tier 1. siRNA / IC50 in DM1-high vs DM1-low thyroid line. Paper 1 in print 이후 collaborator gate.



Paper 11 / 12 cross-reference. 본 atlas 의 36-candidate panel 이 pan-cancer DM1+ lineage 에 portable 한지 (Paper 11), candidate × candidate co-expression network 가 회로 단위로 묶이는지 (Paper 12) 본다.

15. 데이터 / 다운로드 · 자매 페이지

Paper 10 의 figure / table 은 모두 local 산출물에서 재현 가능. Paper 11/12 는 본 atlas 의 36-candidate panel 을 downstream 으로 사용한다.



FIGURES DIR

F01 ~ F17 PNG ·
regenerable



TABLES DIR

T2 ~ T9 source TSVs



PAPER9 FIRST-PASS

upstream data



PDF (THIS PAGE)

Print-friendly export

← PAPER 9 LANDING

SL companion landing
page

← PAPER 9 PLAN

Strategic Ruppín-style
design

← PAPER 9 FIRST-PASS

Local-data
triangulation results



PAPER 11 PANCANCER →

DM1 transfer atlas
across 33 lineages



PAPER 12 NETWORK →

Candidate × candidate
co-expression

← PAPER 1 ANCHOR

DM1 axis 정의



PHASE B PLATFORM

TROP2 ADC carve-
out (non-SL)

← HUB INDEX

8-papers portfolio

Reproducibility. All figures regenerable via `source project/.venv/bin/activate && python project/scripts/p10_atlas_figures.py` ; tables regenerable via `python project/scripts/build_paper10_atlas_html.py` .

What is NOT claimed. No clinical efficacy. No SL-pair claim (deferred). No ISLE/SELECT equivalence (concept-only borrowing). No replacement of driver-mutation profiling. TROP2 explicitly NOT framed as SL.

Scope guard. Voice-protected prose not authored. Paper 1 / 2 / 3 / 4 manuscripts not touched. No new data downloaded. No GPU. No wet-lab.

Live URL. `http://40.82.129.113:8012/papers\_hub\_2026\_05\_04/paper10\_atlas.html`